

MLOps – Assignment 1 Report

Containerised ML Prediction API with CI/CD Pipeline

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Repository: ahmedhanbal/student-ml-api

Course: MLOps

Assignment: No. 1

1. Introduction

This report documents the implementation of Assignment No. 1 for the MLOps course. The assignment repo is hosted at: <https://github.com/ahmedhanbal/student-ml-api>

2. Repository Structure & Branching Strategy

The repository was initialised with a single commit that set up the README, then three feature branches were created and merged sequentially into main through Pull Requests. The final branch graph is shown below.

2.1 Branch Overview

Branch	Purpose	PR #
feature/prediction-api	FastAPI app, Dockerfile, unit tests, CI workflow	#1 & #2
feature/model-metadata	Add model_version field to /health response and update tests	#3
feature/oci-image-metadata	OCI labels in Dockerfile, commit-SHA Docker tags, release workflow update	#4
docs	For maintaining report	#5

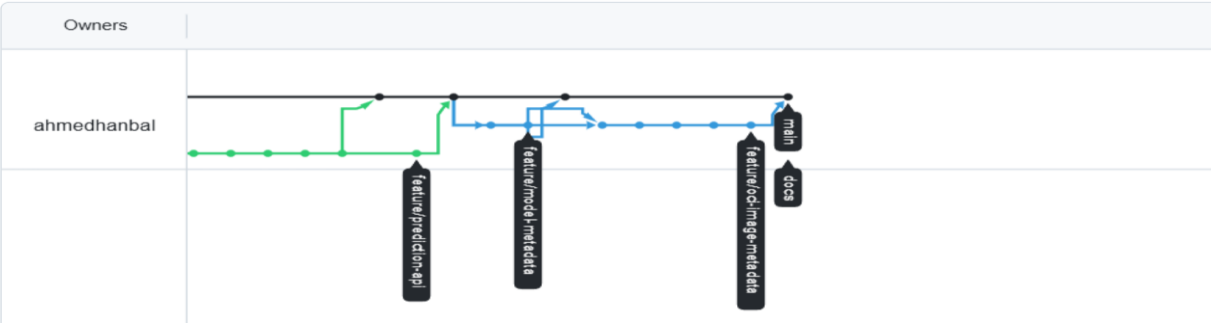
Overview Yours Active Stale All

Q Search branches...

Branch	Updated	Check status	Behind	Ahead	Pull request
main	1 hour ago			Default	
docs	now		0	0	
feature/oci-image-metadata	1 hour ago	✓ 2 / 2	2	0	#4
feature/model-metadata	2 hours ago	✓ 2 / 2	7	0	#3
feature/prediction-api	2 hours ago	✓ 2 / 2	11	0	#2

Network graph

Timeline of the most recent commits to this repository and its network ordered by most recently pushed to and updated daily.



3. Feature Branch 1 – Prediction API (feature/prediction-api)

3.1 What Was Implemented

- Created app.py using FastAPI with two endpoints:
- GET /health – returns status, application name, and version fields.
- POST /predict – accepts a numeric value and returns value × 2 as prediction.
- Created requirements.txt pinning fastapi, uvicorn, pydantic, httpx, and pytest.
- Wrote four pytest unit tests in tests/test_app.py covering health, success, missing input, and invalid input.
- Wrote Dockerfile (python:3.12-slim base) with WORKDIR /app, dependency caching layer, and CMD ["python", "app.py"].
- Added GitHub Actions CI workflow (.github/workflows/ci.yml) triggered on feature/** pushes and main PRs.

3.2 Key Code Snippets

FastAPI application entry points (app.py):

```
@app.get('/health', response_model=HealthResponse)
def health() -> HealthResponse:
    return HealthResponse(status='healthy', application='student-ml-api', ...)

@app.post('/predict', response_model=PredictResponse)
def predict(body: PredictRequest) -> PredictResponse:
    prediction = body.value * 2
    return PredictResponse(input=body.value, prediction=prediction)
```

Dockerfile (dependency-before-app layer ordering):

```
FROM python:3.12-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app.py VERSION ./
EXPOSE 5000
CMD ["python", "app.py"]
```

3.3 Pull Request #1,#2 and CI

Merged

Feature/prediction api #1

ahmedhanbal

merged 7 commits into

main

from

feature/prediction-api

ahmedhanbal

added 7 commits

3 hours ago

feature: add Fastapi prediction and health endpoints

a2e7bae

tests: add unit tests

727ce7a

add dockerfiles

78193e8

workflow: add CI workflows

1a16a65

fix pytest in ci.yml

d32c65f

Introduce a Deliberate Failure

f75732d

tests:remove deliberate failure fix tests

3bdd6bc

ahmedhanbal

merged commit 4fe17f4 into

main

2 hours ago

2 checks passed

View details

Revert

Pull request successfully merged and closed

You're all set — the branch has been merged.

✔ Feature/prediction api CI #4: Pull request #1 opened by ahmedhanbal	feature/prediction-api	Today at 8:34 PM 26s ...
✔ workflow: add release workflow for docker CI #10: Pull request #2 opened by ahmedhanbal	feature/prediction-api	Today at 8:56 PM 29s ...
✔ workflow: add release workflow for docker CI #9: Commit c8fade8 pushed by ahmedhanbal	feature/prediction-api	Today at 8:53 PM 25s ...
✔ Feature/prediction api CI #8: Pull request #1 synchronize by ahmedhanbal	feature/prediction-api	Today at 8:40 PM 25s ...
✔ tests:remove deliberate failure fix tests CI #7: Commit 3bdd6bc pushed by ahmedhanbal	feature/prediction-api	Today at 8:40 PM 29s ...

4. Deliberate Test Failure (Commit f75732d) & Fix (Commit 3bdd6bc)

As part of the assignment requirements, a deliberate test failure was introduced in commit f75732d 'Introduce a Deliberate Failure'. This caused the CI pipeline to fail, demonstrating the safety net provided by automated tests. The failure was then resolved in commit 3bdd6bc 'tests: remove deliberate failure, fix tests', restoring all tests to green.

 Introduce a Deliberate Failure CI #5: Commit f75732d pushed by ahmedhanbal	feature/prediction-api	 Today at 8:38 PM  21s	...
 Feature/prediction api CI #8: Pull request #1 synchronize by ahmedhanbal	feature/prediction-api	 Today at 8:40 PM  25s	...
 tests:remove deliberate failure fix tests CI #7: Commit 3bdd6bc pushed by ahmedhanbal	feature/prediction-api	 Today at 8:40 PM  29s	...
 Merge pull request #2 from ahmedhanbal/feature/prediction-api Release #1: Commit cdfbc55 pushed by ahmedhanbal	v1.0.1	 Today at 8:57 PM  42s	...
 workflow: add release workflow for docker CI #10: Pull request #2 opened by ahmedhanbal	feature/prediction-api	 Today at 8:56 PM  29s	...

5. Feature Branch 2 – Model Metadata (feature/model-metadata)

5.1 What Was Implemented

- Added MODEL_VERSION = 'model-1' constant in app.py.
- Updated the /health endpoint to include model_version in the response payload.
- Updated HealthResponse Pydantic model to include optional model_version and application_version fields.
- Updated test_health_endpoint() to assert on application_version == '1.1.0' and model_version == 'model-1'.
- Merged via Pull Request #3.

5.2 Key Change

```
# app.py
MODEL_VERSION = 'model-1'

payload = {
    'status': 'healthy',
    'application': 'student-ml-api',
    'application_version': '1.1.0',
    'model_version': MODEL_VERSION,
}
```

5.3 Pull Request #3

 **Merged**

Feature: add model metadata #3
ahmedhanbal merged 2 commits into `main` from `feature/model-metadata`

app.py - return updated model metadata
VERSION - change version to v1.1.0



 **ahmedhanbal** added 2 commits 2 hours ago

  `add model metadata` ✗ 7324c7a

  `update test cases for model metadata` ✓ 5a5712e

  **ahmedhanbal** merged commit `1a75502` into `main` 2 hours ago
2 checks passed

[View details](#) [Revert](#)



Pull request successfully merged and closed
You're all set — the `feature/model-metadata` branch can be safely deleted.

[Delete branch](#)

6. Feature Branch 3 – OCI Image Metadata (feature/oci-image-metadata)

6.1 What Was Implemented

- Added OCI-compliant LABEL instructions to the Dockerfile using build ARGs:
 - `org.opencontainers.image.title`, `.version`, `.revision`, `.source`, `.created`
- Updated ci.yml to pass `--build-arg GIT_COMMIT`, `APP_VERSION`, `GIT_REPO`, and `BUILD_DATE` during Docker builds.
- Added release workflow (release.yml) triggered on version tags (`v*.*.*`) that:
 1. Runs pytest.
 2. Logs in to GHCR (GitHub Container Registry).
 3. Builds and pushes images tagged with semantic version, latest, and short commit SHA.
- Added .gitignore entry to exclude local Docker artifacts.
- Merged via Pull Request #4.

6.2 Dockerfile OCI Labels

```
ARG APP_VERSION=1.0.0
ARG GIT_COMMIT=unknown
ARG GIT_REPO=https://github.com/ahmedhanbal/student-ml-api
ARG BUILD_DATE=unknown

LABEL org.opencontainers.image.title="student-ml-api" \
      org.opencontainers.image.version="${APP_VERSION}" \
      org.opencontainers.image.revision="${GIT_COMMIT}" \
      org.opencontainers.image.source="${GIT_REPO}" \
      org.opencontainers.image.created="${BUILD_DATE}"
```

6.3 Release Workflow Summary

The release.yml workflow fires when a Git tag matching `v*.*.*` is pushed. It first runs all tests, then builds the Docker image with full OCI metadata and pushes three tags to `ghcr.io/ahmedhanbal/student-ml-api`: the semantic version (e.g. 1.1.0), latest, and the 7-character commit SHA.

Merged

Feature/oci image metadata #4
ahmedhanbal merged 5 commits into `main` from `feature/oci-image-metadata`

- Images become self-describing: `docker inspect` shows which version and commit produced the artifact.
- `latest` remains a convenience pointer; version tags and commit-SHA tags are the traceable production references.

ahmedhanbal added 5 commits 1 hour ago

Merge pull request #3 from ahmedhanbal/feature/model-metadata

✓ a2a3883

ci: pass OCI image metadata during Docker builds

b956a21

ci: publish Docker images with commit SHA tags

f8fc7ac

docs: explain Docker layer cache ordering

0e25317

chore: add gitignore for local artifacts

✓ baed42e

ahmedhanbal merged commit 9452c65 into main 1 hour ago

2 checks passed

View details

Revert

Pull request successfully merged and closed
You're all set — the `feature/oci-image-metadata` branch can be safely deleted.

Delete branch

OCI Labels

```
$ docker inspect student-ml-api:ci --format "{{json .Config.Labels}}" | python -m json.tool
{
  "org.opencontainers.image.created": "2026-09-10T18:29:47Z",
  "org.opencontainers.image.revision": "9452c65d2d4f813ed3036e0a177674cd532136fa",
  "org.opencontainers.image.source": "https://github.com/ahmedhanbal/student-ml-api",
  "org.opencontainers.image.title": "student-ml-api",
  "org.opencontainers.image.version": "1.1.0"
}
```

Release Workflow and push to GHCR

✅ Merge pull request #3 from ahmedhanbal/feature/model-metadata #2

Summary

All jobs

test

build

Run details

Usage

Workflow file

Annotations

1 warning

build

succeeded 2 hours ago in 23s

Set up job

Checkout repository

Get version

Show version

Login to GHCR

Build Docker image

Push Docker image

Post Login to GHCR

Post Checkout repository

Complete job

GHCR Images

 **student-ml-api** 1.1.0 Public Latest

 Install from the command line

[Learn more about packages](#)

```
$ docker pull ghcr.io/ahmedhanbal/student-ml-api:1.1.0
```

Recent tagged image versions

latest 1.1.0

Published about 2 hours ago · Digest 

 1

1.0.1

Published about 3 hours ago · Digest 

 1

[View and manage all versions](#)

7. CI/CD Pipeline Overview

Two GitHub Actions workflows govern the automation in this project:

Workflow	Trigger	Jobs / Steps
ci.yml (CI)	Push to feature/** OR PR → main	1. Checkout 2. Setup Python 3.12 3. pip install 4. pytest -q 5. docker build (validation)
release.yml (Release)	Push of tag v*.*.*	test: pytest build: docker build with OCI args push: ghcr.io tags (semver, latest, SHA)

Actions

New workflow

All workflows

CI

Release

Management

Caches

Attestations

Runners

Usage metrics

Performance metrics

All workflows

Showing runs from all workflows

Filter workflow runs

18 workflow runs

Workflow Event Status Branch Actor

Feature/oci image metadata

CI #17: Pull request #4 opened by ahmedhanbal

feature/oci-image-metadata

Today at 9:45 PM

28s

...

chore: add gitignore for local artifacts

CI #16: Commit baed42e pushed by ahmedhanbal

feature/oci-image-metadata

Today at 9:42 PM

23s

...

Merge pull request #3 from ahmedhanbal/feature/model-metadata

CI #15: Commit a2a3883 pushed by ahmedhanbal

feature/oci-image-metadata

Today at 9:33 PM

28s

...

Merge pull request #3 from ahmedhanbal/feature/model-metadata

Release #2: Commit 1a75502 pushed by ahmedhanbal

v1.1.0

Today at 9:16 PM

41s

...

Feature: add model metadata

CI #14: Pull request #3 synchronize by ahmedhanbal

feature/model-metadata

Today at 9:14 PM

25s

...

8. Unit Testing

All tests are located in tests/test_app.py and use FastAPI's built-in TestClient.

Test Function	What It Verifies
test_health_endpoint()	GET /health returns 200, status='healthy', correct app & model versions
test_predict_success()	POST /predict with value=10 returns prediction=20
test_predict_missing_input()	POST /predict with empty body returns 422 Unprocessable Entity
test_predict_invalid_input()	POST /predict with string value returns 422 Unprocessable Entity

```
$ python -m pytest --disable-warnings -v
===== test session starts =====
platform win32 -- Python 3.13.14, pytest-8.3.5, pluggy-1.6.0 -- C:\Users\ahmed\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\ahmed\code\mlops\ai
plugins: anyio-4.15.1
collected 4 items

tests/test_app.py::test_health_endpoint PASSED [ 25%]
tests/test_app.py::test_predict_success PASSED [ 50%]
tests/test_app.py::test_predict_missing_input PASSED [ 75%]
tests/test_app.py::test_predict_invalid_input PASSED [100%]

===== 4 passed, 1 warning in 0.35s =====
```

9. Extra

I used cursor agent to debug pushing docker image to GHCR that is why it shows in contributors. All the code part including CI,CD was done by myself.